

Omar Mohammad Shaker

Backend Software Engineer

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About

Top-ranked CS student at GIU (0.7 GPA / A+, Excellence Scholarship) specializing in backend development with Python, Django, and REST APIs — with growing depth in Cloud and AI. Built projects from e-commerce platforms to ML models from mathematical first principles, and run a daily “Python in 60 Seconds” LinkedIn series. Seeking a Software Engineer / Backend Engineer internship to do deep technical work on real-world systems.

Education

German International University (GIU)

2025 – 2029 (Expected)

Bachelor of Computer Science & Informatics

Cairo, Egypt

- **GPA:** 0.7 (German Scale, Equivalent to 4.0 / A+) | **Rank:** 3rd / 500+ | **Excellence Award Scholarship**
- **Coursework:** DSA, Java, Computer Organization, Digital Logic Design, OOP, Theoretical CS, Networks, Linear Algebra

Skills

- **Languages & Frameworks:** Python, Django, Java, SQL (PostgreSQL, Microsoft SQL Server), Assembly (MIPS)
- **Tools & Technologies:** REST APIs, Git, GitHub, AWS, Jupyter, Eclipse, VS Code
- **Machine Learning:** Linear Regression (Normal Equation, Gradient Descent, Ridge), Model Evaluation (MSE, R^2)
- **CS Fundamentals:** Data Structures & Algorithms (Backtracking, DP, Graphs, Trees), OOP, Linear Algebra

Experience

Independent Backend & Software Engineering Projects

Jan 2024 – Present

Personal Portfolio

- Designed and developed **5+ production-style** backend, ML, and GUI systems using Python, Django, Java, and SQL
- Implemented RESTful APIs, authentication, and role-based access applying **DSA/OOP** principles

AWS Cloud Club — GIU

Nov 2025 – Present

Logistics Lead & Core Team Member

- Co-organized **4 sessions + 3 workshops + 2 events at GIU and AUC**, reaching **300+ students**
- Sourced and onboarded **3+ field experts** for technical talks, coordinating both online and offline sessions

Technical Content Creator — LinkedIn

Dec 2025 – Present

“Python in 60 Seconds”

- Published daily Python content simplifying complex concepts
- Grew audience by **350% followers in 50 days**, and published **60+ posts**, maintained daily streak for **3+ months**

Technical Interview Prep — LeetCode | Ongoing

- Solved **110+ problems** across trees, graphs, DP, backtracking, and databases

Projects

Store Hub | Python, Django, SQL

- Built a full-stack e-commerce platform across **7 Django apps with 30+ views, 12+ database models**, and a merchant dashboard covering product, order, and inventory management
- Designed a normalized relational schema and implemented RESTful backend logic with authentication, session management, and role-based access across the full user-merchant flow

DooR DasH Board Game | Java, JavaFX, OOP

- Engineered a 2-player, 100-cell strategy board game across **20+ Java classes** with **10+ distinct cell effect types**, leveraging inheritance, polymorphism and abstraction for a fully event-driven JavaFX GUI

Machine Learning: Linear Regression | Python, NumPy, Pandas, Scikit-learn, Jupyter

- Developed Linear Regression from scratch using matrix calculus $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$, achieving machine-precision agreement with scikit-learn (**<1e-13 error**)
- Implemented Gradient Descent with convergence to optimal weights ($\Delta < 1e-3$) and evaluated performance using **MSE and R^2 metrics ($\approx 913 / 0.734$)**
- Extended model with Ridge Regularization, experimentally analyzing bias-variance tradeoff across λ values (**0 to 100**)

Certifications

- Master the Coding Interview: Data Structures + Algorithms | Udemy
- SQL Full Course From Zero to Hero | Data with Baraa
- AWS AI Practitioner Challenge | Udacity (In Progress)
- McKinsey Forward Program | McKinsey Academy (In Progress)